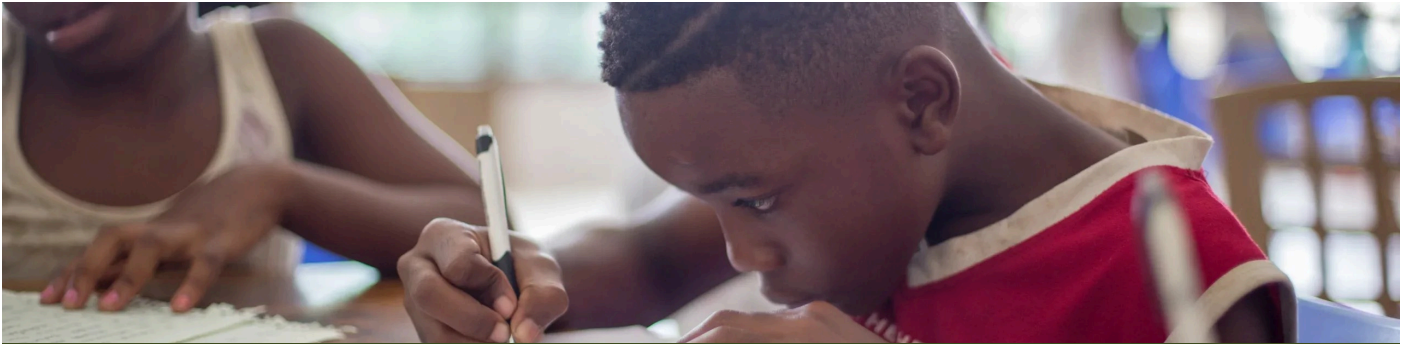


40 | OVERVIEW



Climate, Air Quality, and the Developing Brain: Protecting Youth Health and Cognition

Abstract

Climate change brings hotter temperatures and worse air pollution, and both can affect how children's brains develop and how well students learn. This lesson, based on a Harvard Chan C-CHANGE guide, reviews studies linking particle pollution to lower IQ scores and heat to lower test scores, and explores steps families and schools can take to protect young minds.

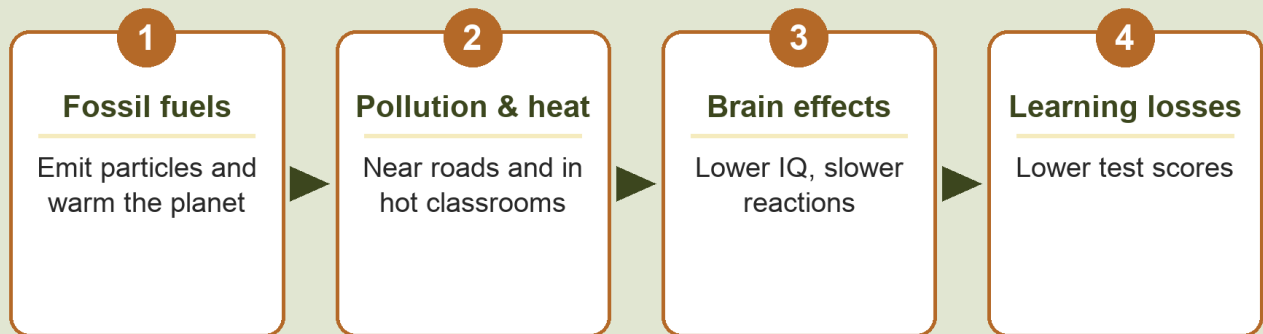
Introduction

Most people think of air pollution as a lung problem and heat as a comfort problem. But growing evidence suggests both can affect the brain. Burning fossil fuels in power plants, cars, and trucks produces carbon pollution that drives climate change, and it also creates particulate matter (PM) that can harm developing brains. Meanwhile, climate change has already made temperatures hotter and heat waves more frequent and intense across most of the United States. Studies suggest that heat and air pollution can influence everything from how a brain develops to how well a student performs on a test, with some students affected more than others.

How Do Scientists Study This?

Researchers use observational studies to compare groups of young people with different exposures. In Los Angeles, scientists compared the IQ scores of 9-to-11-year-olds and 18-to-20-year-olds exposed to different levels of particulate matter. In Massachusetts, researchers compared children living within 50 meters of a major roadway, where traffic pollution is higher, with children living more than 200 meters away. To study heat, researchers analyzed the test scores of millions of U.S. high school students after hotter and cooler school years, and compared college students in dorms with and without air conditioning during a summer heat wave.

How climate can affect learning



Measure exposure

Scientists estimate how much pollution or heat young people experience where they live and learn.

Measure thinking

IQ tests, coordination tasks, test scores, and reaction times measure cognition.

Compare groups

Results for more-exposed and less-exposed groups reveal links to pollution and heat.

What Did They Discover?

In Los Angeles, young people exposed to more particulate matter had lower IQ scores. In Massachusetts, children living within 50 meters of a major road had lower IQs and poorer hand-eye coordination than children living more than 200 meters away. For heat, high school students scored lower after a hot school year than after a cool one. Low-income and minority students were affected more, and air conditioning in schools almost completely eliminated the effect. College students in dorms without air conditioning during a heat wave had lower test scores and slower reaction times. A 2023 EPA report projected that climate-driven heat could reduce academic achievement by 4% to 7% each year, potentially affecting future income.

Heat, pollution, and learning

50 m

from a major road: lower IQ than kids 200 m away

4–7%

projected yearly drop in achievement from heat

A/C

in schools almost erased heat's effect on test scores

Scientists still need to learn much more about how hot is too hot for children to think clearly and learn.

Takeaways

Pollution affects the brain

More particle pollution is linked to lower IQ and poorer coordination.

Heat affects learning

Hot school years and hot dorms are linked to lower scores and slower reactions.

Cooling is fairness

Air conditioning in schools nearly erased heat's effect, which hit some students harder.

Key Terms

Cognition

Mental processes such as thinking, learning, remembering, and paying attention.

Particulate matter (PM)

Tiny airborne particles from sources like traffic that can reach the lungs, bloodstream, and brain.

Observational study

Research that compares groups as they naturally are, without assigning treatments.

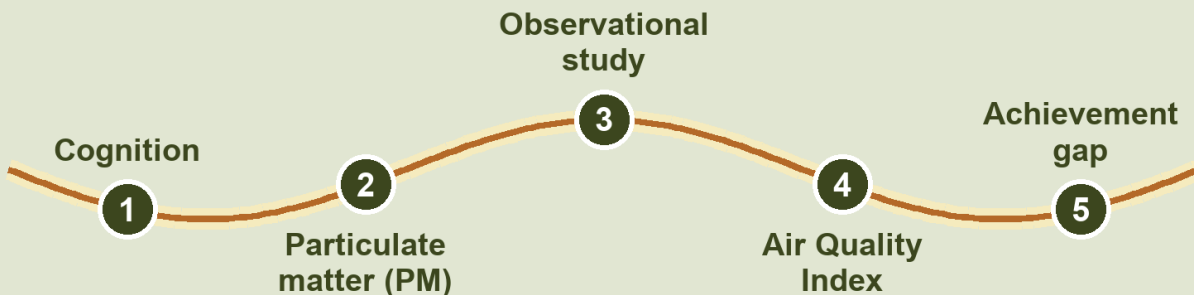
Air Quality Index

A color-coded scale, used by tools like airnow.gov, that shows how healthy the air is to breathe.

Achievement gap

Differences in academic performance between groups of students, which heat can widen.

Vocabulary Path · Climate, Air Quality, and the Developing Brain



Activity

Is Our School Climate-Ready?

Students audit their school for heat and air pollution risks and propose improvements.

Materials: School audit checklist; Thermometer (optional); Air quality website such as airnow.gov

1. Check today's Air Quality Index for your area and record its color and meaning.
2. Survey classrooms: which have air conditioning, fans, or shade? Record temperatures if possible.
3. Identify pollution sources near school, such as busy roads or idling buses.
4. Using evidence from the lesson, rank the top two risks to students' learning.
5. Propose one policy, such as a no-idling rule or cooling plan, and explain how it would help all students fairly.

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Climate, Kids, and Health: Brain Development. <https://hsph.harvard.edu/research/climate-health-c-change/climate-kids-and-health/brain-development/>